

GRAD- E1287: Emissions Pricing**1. General information**

Course Format	Onsite
Instructor(s)	Christian Flachsland (CF), Lion Hirth (LH)
Instructor's E-mail	flachsland@hertie-school.org , hirth@hertie-school.org
Assistant (if applicable)	to CF: Name: Alwine Hoppe Email: hoppe@hertie-school.org to LH: Name: Alex Karras Email: karras@hertie-school.org
Instructor's Office Hours	Will be announced in class

Link to [Study, Examination and Admission Rules and MIA, MDS and MPP Module Handbooks](#)

For information on **course rooms, times and session dates**, please consult the [Course Schedule](#) on *MyStudies*.

Instructor Information:

Christian Flachsland is Professor of Climate Policy at the Hertie School and Director of the School's Centre for Sustainability. He is also a Research Fellow at the Mercator Research Institute on Global Commons and Climate Change (MCC), where he previously led the working group 'Governance'. His research focuses on the design, governance, and politics of climate, energy, and sustainability policy. He publishes in *Science*, *Nature Climate Change*, and leading journals on climate policy and climate politics. He co-coordinates the research on Governance in the Kopernikus-Ariadne project, a major research consortium assessing climate policy options for Germany and Europe, funded by the German Federal Ministry of Education and Research (BMBF).

Lion Hirth is Professor of Energy Policy at the Hertie School. His research interests lie in the economics of wind and solar power, energy policy instruments, and electricity market design, as well as open data and open-source modelling. Lion is also a Research Fellow at the Mercator Research Institute on Global Commons and Climate Change (MCC), Director of the consulting firm Neon, and Secretary of Strommarkttreffen, a network of energy professionals. His academic publications on the market value of wind energy and on the integration of renewable energy into power systems have won several

awards. He holds a PhD in Energy Economics, and Diploma in Economics, and a Magister in Political Science.

2. Course Contents and Learning Objectives

Course contents:

The course in a tweet: This is an advanced course on emissions trading and taxation. It covers conceptual basics, design options, price formation in cap-and-trade schemes. We also discuss the many pitfalls when designing robust emission pricing systems: carbon leakage, linking, winners and losers, politics.

The significance of emissions pricing. Climate change is caused by greenhouse gases (carbon dioxide, methane, and others) being emitted into the atmosphere. The fact that such emissions are not priced is “the greatest market failure that the world has seen”, as Lord Nicholas Stern put it. In addition, emissions of local pollutants (SO_x, NO_x, particular matter) rank high on the list of most pressing policy problems in many parts of the world, in particular in many of the megacities in fast-growing developing and emerging economies. Economists have long advocated putting a price tag on emissions as an efficient means to address emissions. However, putting emissions pricing into practice has turned out to be difficult. The purpose of this class is to understand why, and how robust emission pricing policies ought to be designed.

Topics. First, students will be introduced to the environmental economic theory of externalities, such as climate change. We will discuss the two principal options for pricing externalities, taxation and cap-and-trade. Then, we will turn to specific topics of real-world emissions pricing: the intertemporal nature of trading schemes that requires governments to earn investors’ trust, the political economy of initial allowance allocation and revenue recycling, international and inter-sectoral carbon leakage, and the (im)possibility to implement sensible national emission policies for countries that are part of international trading schemes. During the last third of the course, we will study concrete emission pricing policies, with the EU Emission Trading Scheme being our main case study throughout.

Learning objectives:

This class provides a comprehensive overview of the theory and, more importantly, the political practice of pricing emissions. This is an advanced course that provides a deep-dive in a narrow, but technical and complex topic. The objective of this class is to equip students with the analytical and empirical understanding needed to implement, assess and advance carbon pricing schemes in governments, companies, and civil society.

Target group:

This is an advanced course. It is designed for students with significant prior knowledge in climate change, climate policy, and environmental economics. It is *not* an introduction to climate change or environmental policy more broadly. Rather, we will narrowly focus on emissions pricing. Working at the research frontier, we will read mostly research papers.

Prerequisites:

Prerequisites. Students are expected to have solid prior understanding of the following issues:

- Climate change (i.e., the physics of the greenhouse effects, consequences of climate change)
- Environmental economic concepts (e.g., public goods and externalities, principal approaches to internalize externalities)
- Climate policy (e.g., a rough understanding of the EU Emissions Trading Scheme, carbon taxes, the Paris agreement)

Students who do not bring this experience are expected to do significant reading ahead of the first session the following documents:

- Wikipedia: [Global Warming](#), [Greenhouse Effect](#), [Radiative Forcing](#), [Carbon Cycle](#)
- [NRC \(2012\): Climate Change](#)
- [Carbon Brief \(2018\): Climate Sensitivity](#)
- [UNEP \(2018\): Emissions Gap Report](#)
- [Stern Review \(2006\): Executive Summary](#)
- [Bernauer \(2013\): Climate Change Politics](#)

Diversity Statement:

Understanding and respect for all cultures and ethnicities is central to the teaching at Hertie. Being mindful of diversity is an important issue for policy professionals in the planning, implementation, and evaluation of programmes designed for specific groups, populations, or communities. Diversity and cultural awareness will be integrated in the course content whenever possible.

3. Grading and Assignments

Policy on the use of AI tools for assignments

The use of generative AI tools (such as ChatGPT) is **permitted throughout this course**. Students must adhere to the specific instructions provided for each assignment. Where AI tools are used, students are expected to clearly indicate how they were used and to ensure the originality and academic integrity of their work. **Students must include a Statement of Authorship in all written assignments.** Improper or undisclosed use of AI tools may be considered a violation of academic integrity policies. Please consult the course instructor in case of doubt. Further guidance is available in the [Artificial Intelligence Tools at the Hertie School - Teaching Guidelines for Instructors and Students](#), including a template for the **Statement of Authorship**, under section 1.1.

Composition of Final Grade:

Assignment 1: Mid-term exam		Closed-book in-class exam of 60 min.	35%
Assignment 2: Debate	Session 8		10%
Assignment 3: Paper outline	Deadline: at least two weeks prior to the presentation	Bring to meeting with Christian Flachsland	5%
Assignment 4: Group presentation	Sessions 10 and 11		10%
Assignment 5: Group paper with individual sections	Deadline: 22 December, 18:00 h	Submit via Moodle	40%

Assignment Details

Mid-term exam. The exam is a closed-book in-class exam of 60 min. The exam will include both essay and analytical questions.

Debate. Students will be allocated to one of the three groups. The debate will take place in session 8. Unless student performance is remarkably different, each group will receive a joint presentation grade.

Group presentations. Students will choose group presentation topics from case studies suggested by the instructors. Allocation of students to their topic will take place early in the semester to enable careful preparation. Presentation topics are identical to the final paper topic. The presentations will take place Sessions 10 and 11. Unless student performance is remarkably different, each group will receive a joint presentation grade.

Outline. Each group of students is required to consult with Christian Flachslund the orientation and outline of their presentation at least two weeks prior to the presentation. Students are responsible for scheduling this meeting well in advance. Students are expected to bring a 2-page outline of the presentation/paper to the meeting. Details on scheduling the meeting will follow in class.

Group paper with individual sections. Student groups (max. 4 students per team) write a policy paper with max. 3.000 words per student (e.g. a group of three students would write a 9.000 words paper). The policy paper topics and questions addressed are largely identical with those of the presentations. The mandatory meetings for preparing presentations can also be used for discussing the paper outline. Students can opt for receiving one joint grade, or clearly indicate which student wrote which distinguishable part of the joint paper to receive separate grades. Submission of the policy paper via Moodle by 21 May 18:00h.

Late submission of assignments: For each day the assignment is turned in late, the grade will be reduced by 10% (e.g. submission two days after the deadline would result in 20% grade deduction).

Attendance: Students are expected to be present and prepared for each class session. Active participation during lectures and seminar discussions is essential. Please note that students can miss up to two sessions (out of twelve) if no course assignments are affected. **Students do not need to inform the course instructor(s) in advance if they are unable to attend a session.** For further information please consult the [Examination Rules](#) §10.

Academic Integrity: The Hertie School is committed to the standards of good academic and ethical conduct. Any violation of these standards shall be subject to disciplinary action. Plagiarism, misuse of AI, free riding in group work, and other deceitful actions are not tolerated. See [Examination Rules](#) §16, the Hertie [Plagiarism Policy](#), and [the Hertie Guidelines for Artificial Intelligence Tools](#).

Compensation for Disadvantages: If a student furnishes evidence that they are not able to take an examination as required in whole or in part due to disability or permanent illness, the Examination Committee may upon written request approve learning accommodation(s). In this respect, the submission of adequate certificates may be required. See [Examination Rules](#) §14.

Extenuating circumstances: An extension can be granted due to extenuating circumstances (i.e., for reasons like illness, personal loss or hardship, or caring duties). In such cases, please contact the course instructor and Examination Office *in advance* of the assignment deadline.

4. Course Sessions and Readings

All course readings can be accessed on the course Moodle page.

Please note that fall semester Moodle course pages are available until 31 March of the following semester. Spring semester Moodle courses are available until 31 September of the following semester. After that, in accordance with German copyright law, they cannot be accessed.

Session 1: The EU ETS	
The State of Carbon Pricing in Europe	
Required Readings	ICAP Brief #1 What Is Roser (2021): The argument for a carbon price Ellerman et al. (2016): 10 years EU ETS Baranzini et al. (2017): Carbon pricing CLEW (2022): Germany's carbon pricing
Optional Readings	RFF (2018): Carbon Pricing 101

Session 2: Externalities and Public Goods	
Required Readings	Keohane & Olmstead (2016), ch. 5 Perman et al. (2012), ch. 4.10, 5.1-5.6 The Economist (2017): Pigouvian taxes Howard & Sterner (2025): Climate damages
Optional Readings	Coady et al. (2016)

Session 3: Cap-and-trade vs. Taxation	
Required Readings	ICAP Brief #2 Arguments and #8 Tax Keohane & Olmstead (2016), ch. 8 Hepburn (2009), ch. 18 Meckling & Hepburn (2013), p. 475-480 Hayek (1945): Use of knowledge
Optional Readings	Stavins (2022) Goulder and Parry (2008) Keohane & Olmstead (2016), ch. 9, 10 Fankhauser & Hepburn (2010): Space

Session 4: Emissions Pricing in Practice	
Required Readings	ICAP Brief 3: Around the World Richie & Rosado (2022) World Bank (latest) Carbon Pricing , ExSum,1,2 ICAP & PMR (2016) ETS Handbook, p. 5-28 Schmalensee and Stavins (2017)
Optional Readings	Andersson (2019) Dechezlepretre et al. 2022

Session 5: Value Allocation	
Required Readings	ICAP Brief 5: Distribution and Brief 6 Revenues RFF (2019): Revenue Use Options

	RFF (2020): Effects across Income Groups Zetterberg et al. (2012) Klenert et al. (2018)
Optional Readings	Hirth & Ueckerdt (2013) Goulder (2013): Tax system

Session 6: Carbon Leakage

Required Readings	Jakob (2021): Climate policy and trade PMR (2015), Ex Sum ICAP (2020): Carbon leakage , sec 3,5 Killick et al. (2021): CBAM proposal
Optional Readings	Dechezleprêtre & Sato (2017) Martin et al. (2014) , p. 2482-2497 OECD (2015) ; Branger & Quirion ('14)

Session 7: Time and Credibility

Required Readings	ICAP Brief 8: Banking & Borrowing Carbon Brief (2020): When 1.5° warming? Fuss et al. (2018) Acworth et al. (2017), SPM Fankhauser & Hepburn (2010): Time , sec. 1-2
Optional Readings	Newell et al. (2005) Kollenberg & Taschini (2016) Rubin (1996), sec. 1-2

Session 8: Debate: Evaluating Carbon Pricing Policies

Required Readings	Stokes & Mildenberger 2020 Roberts 2021 Patt and Lilliestam 2018 Lilliestam et al. 2020 van den Bergh & Savin 2021
Optional Readings	-

Session 9: Linking Trading Schemes

Market Stability Reserve

Required Readings	ICAP Brief #4: Linking ETS Ranson & Stavins (2015) Flachsland et al. (2009): To link Perino (2018): Puncture Waterbed Perino (2021): ETS Autopilot
Optional Readings	Pizer & Yates (2015) Osorio et al. (2021): MSR

Session 10: Case study 1

Case study 2	
Required Readings	(TBA, depends on choice of students)
Optional Readings	(TBA, depends on choice of students)

Session 11: Case study 3 Case study 4	
Required Readings	(TBA, depends on choice of students)
Optional Readings	(TBA, depends on choice of students)

Session 12: The Future of Carbon Pricing in Germany & EU	
Required Readings	TBA RFF (2020): Transport Sector
Optional Readings	TBA